



Network Swiss Army Knife (NSAK)

Bachelor Thesis – Integrating Agentic AI for Network Reconnaissance

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Lukas von Allmen and Frank Gauss | Bern University of Applied Sciences

Overview

- ▶ Introduction
- ▶ Implementation
- ▶ Evaluation Setup
- ▶ AI Integration
- ▶ Results
- ▶ Discussion and Conclusion
- ▶ Demo: Agentic Reconnaissance in Action

Introduction

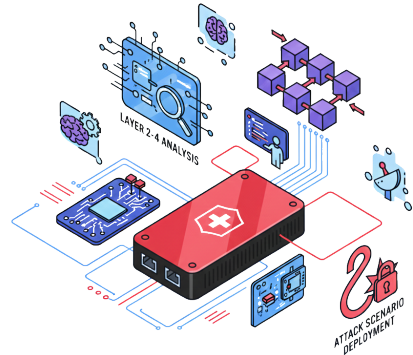
Starting Point: The NSAK Framework

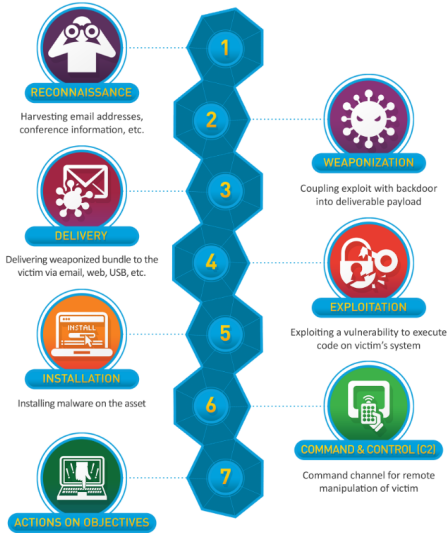
Established in the predecessor project (Project 2)

A modular, open-source network security framework

Built around four reusable resource types:

-  **Device** *execution host at a network position*
-  **Environment** *target network topology*
-  **Scenario** *a red/blue team use case*
-  **Drill** *reusable building block*





- Gather baseline data
- Detect
- Alert
- Investigate
- Plan a response
- Execute

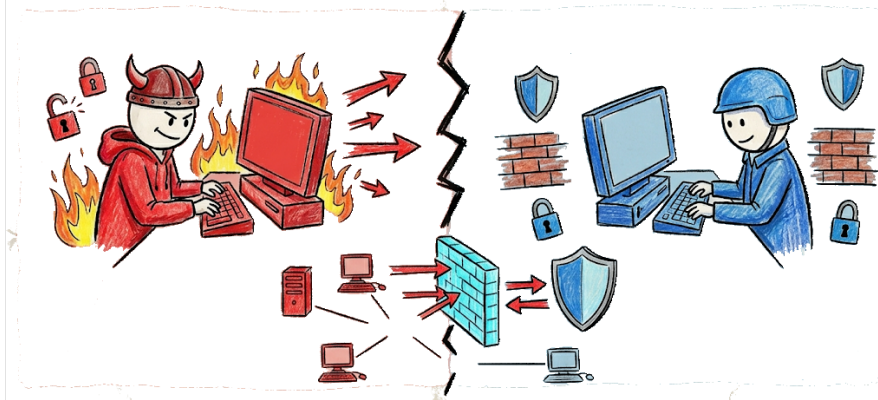


Figure: Red and Blue Team

Reconnaissance

The first stage of the cyber kill chain



Gathering information about a target network *before* any attack – it builds the map every later stage relies on, and is the focus of this thesis.

-  **Host discovery** *which systems are reachable on the network*
-  **Service & port enumeration** *what is running and exposed*
-  **Characterization** *OS, versions and vulnerability indicators*

Hypothesis and Research Questions

From static scenarios to agentic AI

Hypothesis

Integrating **agentic AI** into NSAK improves **adaptability, flexibility** and **automation** over static scenarios, while **lowering the operator's required expertise** to interpret results and plan follow-up steps.

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Integrating **agentic AI** into NSAK improves **adaptability, flexibility** and **automation** over static scenarios, while **lowering the operator's required expertise** to interpret results and plan follow-up steps.

- RQ1** How does an agentic AI scenario perform **compared to a static reference** scenario in network reconnaissance?
- RQ2** To what extent can agentic AI **support operators** during interactive reconnaissance?
- RQ3** To what extent is the implementation **suitable for real-world** deployment?

Project Goals

Order of implementation: Must → Should → Could

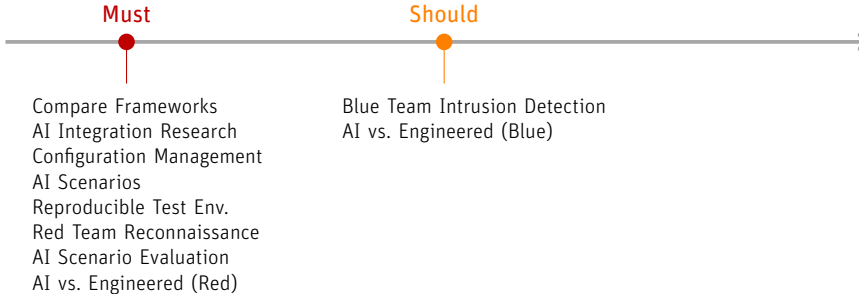
Must



Compare Frameworks
AI Integration Research
Configuration Management
AI Scenarios
Reproducible Test Env.
Red Team Reconnaissance
AI Scenario Evaluation
AI vs. Engineered (Red)

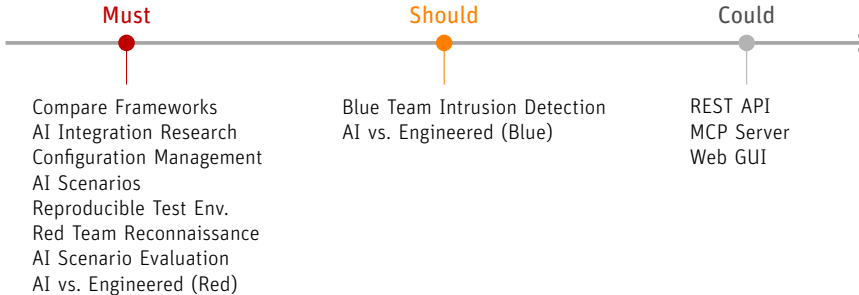
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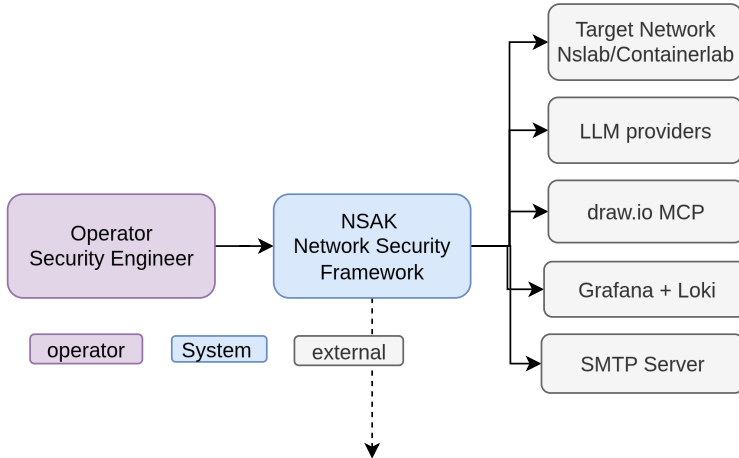
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Implementation

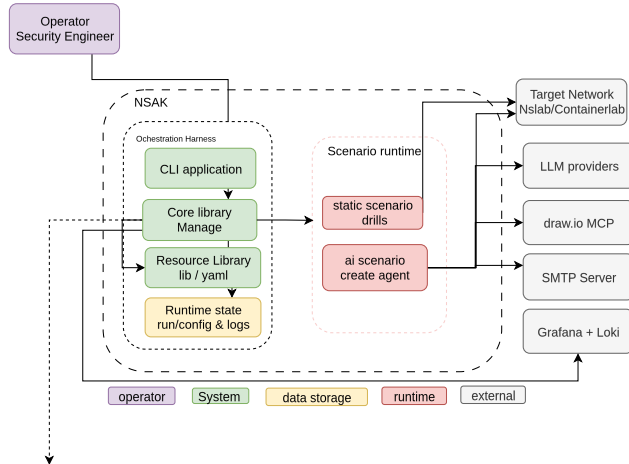
Layered Architecture

system-diagram



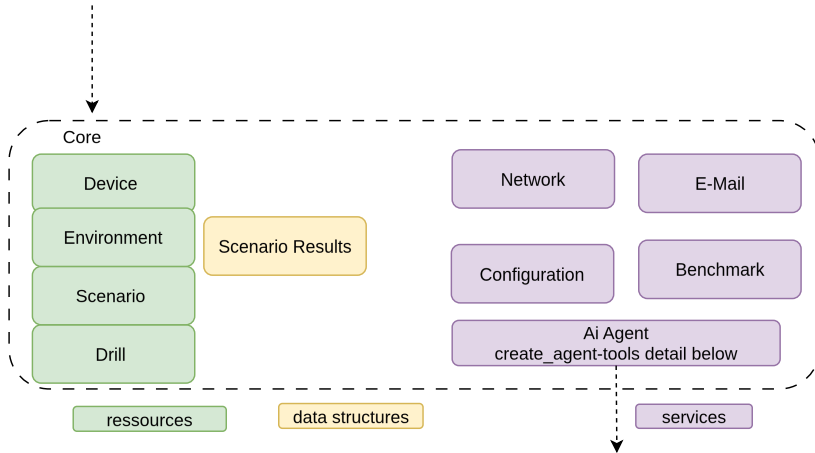
Layered Architecture

Lean Component Diagram



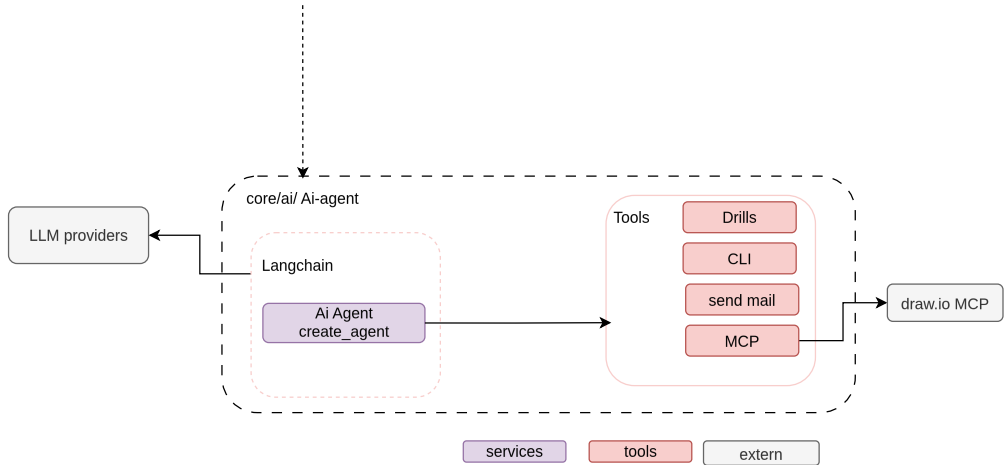
Layered Architecture

Core Diagram and Agent



Layered Architecture

Core Diagram and Agent



Evaluation Setup

Methology

3 LLMs • 2 Network Environments • Static Reference Baseline

LLMs

- qwen3: Self-hosted via Ollama
- gpt-oss: BFH-TI MLMP
- claude-opus-4-7: Anthropic API

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Network Environments

- Reproducible Containerlab Network
- BFH Network Security LAB

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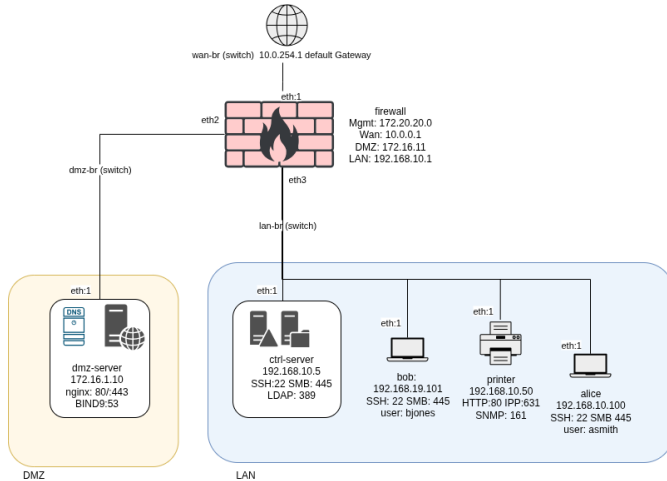
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Reconnaissance Scenario

- Accepts an interface and/or a subnet as arguments
- discover_network_interfaces:
Iterates all interfaces if none is provided
- dhcp_request: Gets an IP via DHCP if needed
- discover_hosts: arp-scan & nmap
- enumerate_services: nmap NSE scripts
- Returns a ReconnaissanceScenarioResult

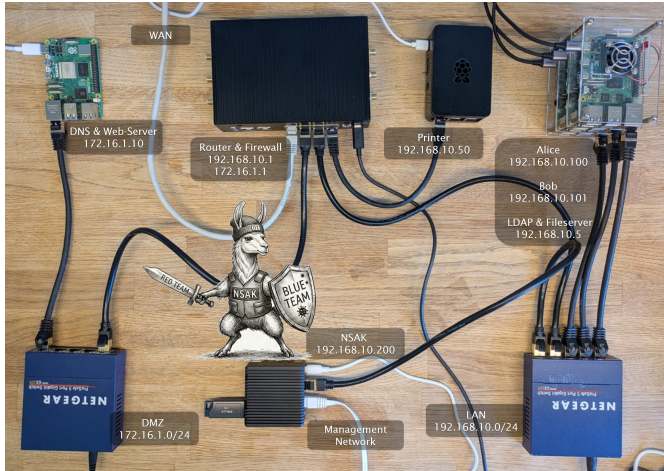
Reproducible Test Environment

A virtual enterprise network built with Containerlab



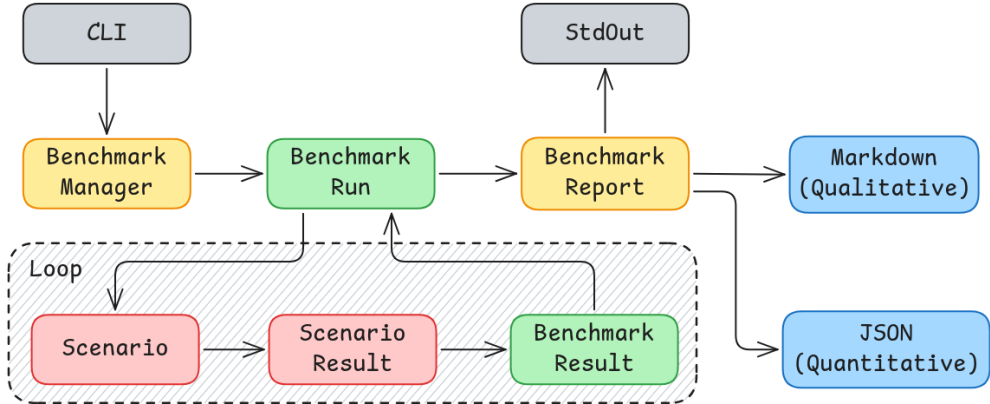
Physical Lab

The same network implemented with real hardware



Benchmarking Suite

Automated test runs for non-interactive scenarios



AI Integration

1. Provider Model und Agent
2. Tool calling (Langchain Tools) (MCP) (human interaction hook) (cli) (draw io) (send-mail)
3. 1. Single Agent interactive unstructured output
4. 2. Single Agent Reconnaissance unstructured output
5. 3. POC Blueteam intrusion detection - not followed time and scope - unstructured output
6. 4. Single Agent reconnaissance structure output (context-window, grammar -> why structured output create larger grammar)
7. 5. Multi Agent reconnaissance structured output (issue with smaller models) () Containerlab evaluation smaller model
8. 6. Multi Agent reconnaissance unstructured output

Concept 1: Taming the Context

From one overloaded agent to a focused pipeline

The problem

- One agent does everything in a single long conversation
- Every scan output piles up → the **context keeps growing**
- Smaller models lose the thread and start failing
- Cost and runtime become hard to predict

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- Each phase is a **fresh agent** with its own, small context
- We pass on only the **distilled result**, not the whole history

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 Trade-off: Improves results when the context window is almost full, but adds orchestration overhead.

Concept 2: Keeping the Operator in Control

The agent proposes – the human decides on risky actions

The problem

- An autonomous agent can run **any** command or send reports on its own
- In a security context that is unacceptable without oversight

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- The agent **pauses** before sensitive actions and asks for a decision
- Shell commands → **approve** before they run
- ✉ Sending a report → **approve / reject**
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↺ After the decision the agent resumes exactly where it stopped – no restart, no lost progress.

Single Agent

- One prompt to achieve the goal
- Long chain of tool calls, **growing context**
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Multi Agent *divide-and-conquer*

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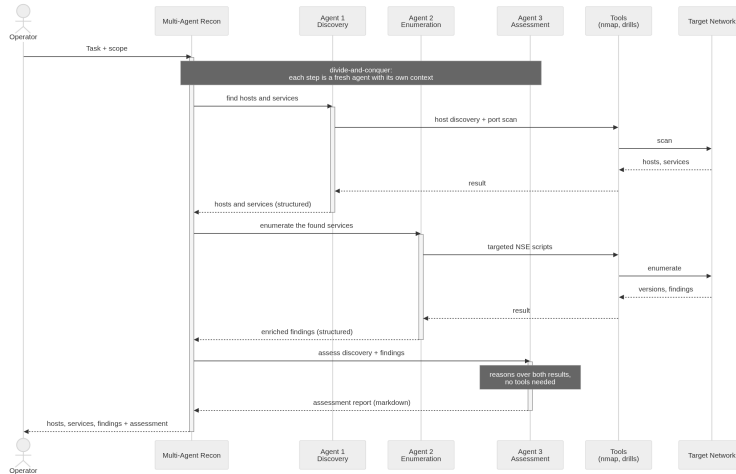
Structured vs. Unstructured output

Structured (schema-constrained): great for automated metrics, but **high failure rate** – especially for small models.

Unstructured: needed in the BFH lab where all models failed to produce valid structured output; freed the agent but required manual evaluation.

Multi-Agent Reconnaissance Pipeline

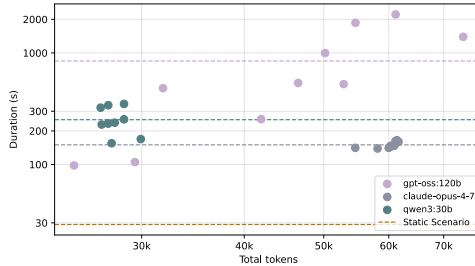
Discovery → enumeration → assessment, each a fresh agent



Results

Containerlab – Multi-Agent vs Single-Agent

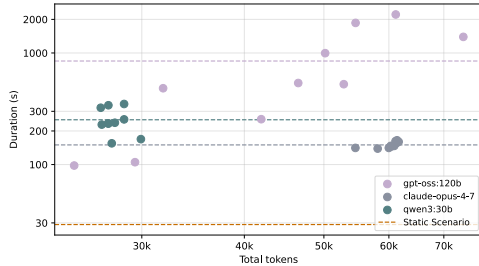
Tokens vs. duration – multi-agent first, then single-agent



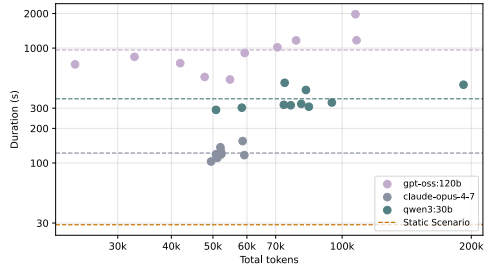
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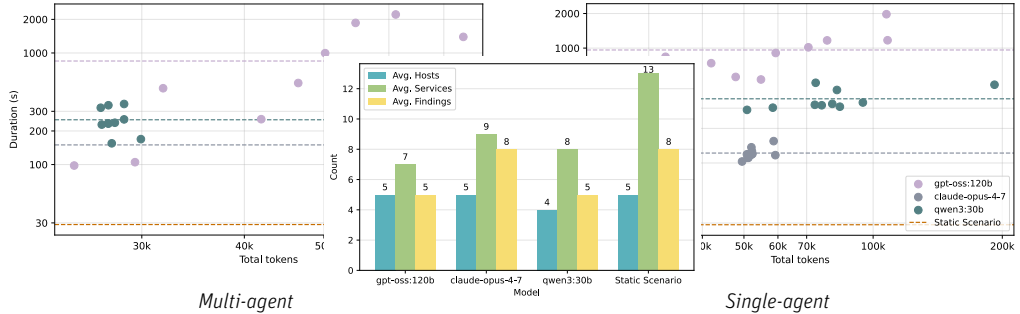
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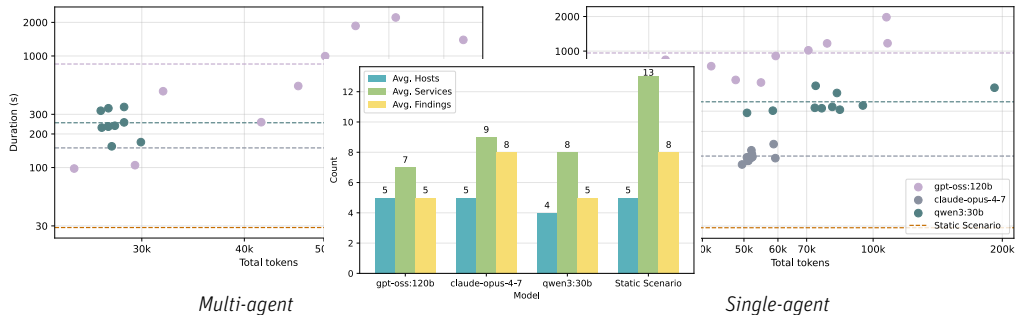
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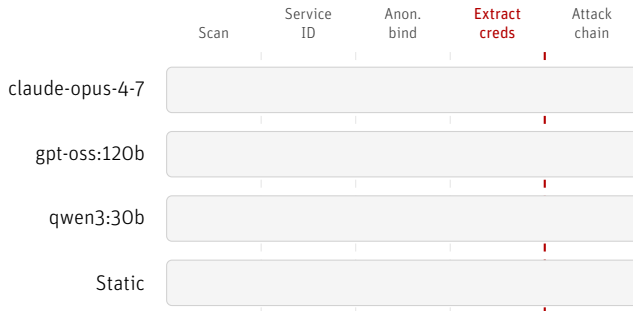
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- Decomposition: **qwen3:30b** drops tokens **–68%** and duration **–30%**.
- **claude-opus-4-7** keeps identical discovery quality (slight +13% token overhead).

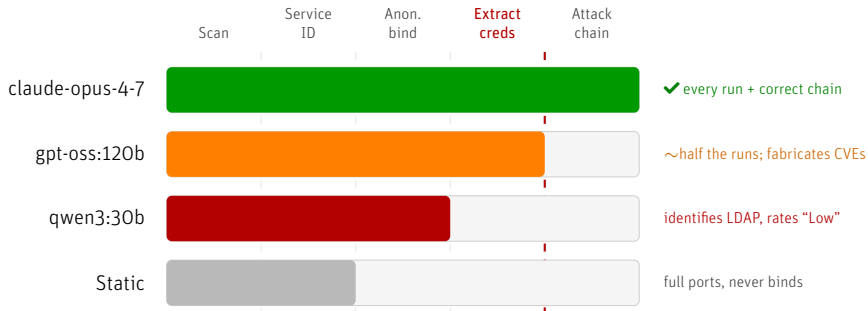
Qualitative Results + Interactive Mode

Q How far does each actor get along the attack chain?



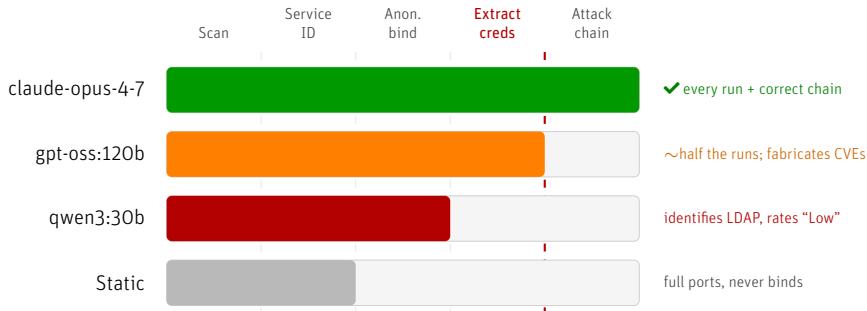
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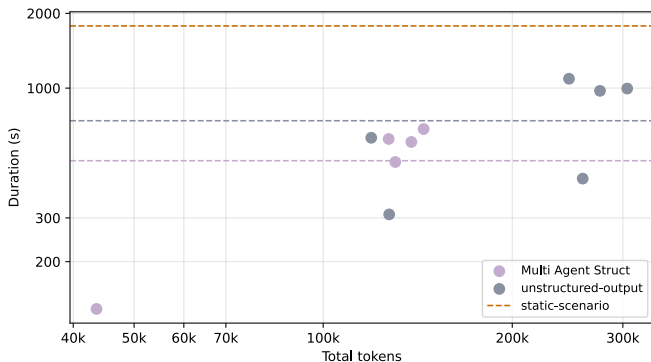
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Trade-off: static = breadth & determinism; AI = interpretive depth (severity-rated assessment, next steps).

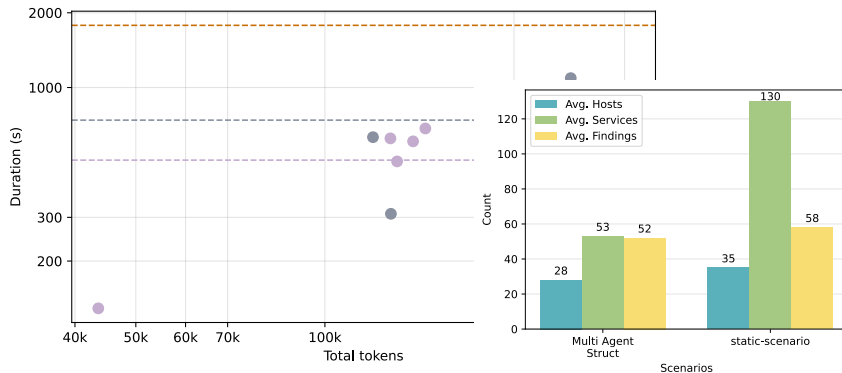
Network Security Lab – Multi-Agent Benchmark

NSlab - Findings



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Discussion and Conclusion

Key Findings

Takeaways across both environments

✓ What works

- Even smaller models can autonomously run **standard red-team reconnaissance**
- **Multi-agent decomposition** helps smaller models (−68% tokens for **qwen3:30b**)
- AI delivers **depth**: assessment, prioritized findings, remediation

Key Findings

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- Even smaller models can autonomously run **standard red-team reconnaissance**
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⚠ What limits it

- **High failure rate** – best-case subset
- **Hallucinations**: fabricated CVEs, honeypots

Answering the Research Questions

RQ1 AI vs. static: Same discovery power as baseline, no gains in speed/determinism. Adds value via **reasoning and interpretation** (credentials, banner reasoning, severity scoring).

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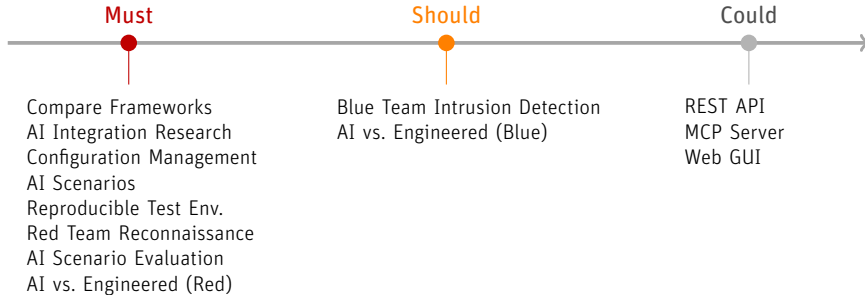
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- RQ2 Operator support:** Converts scans into structured, severity-ranked output + next steps. **--interactive**
Operator stays in the loop and control.

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- RQ2 Operator support:** Converts scans into structured, severity-ranked output + next steps. **--interactive**
Operator stays in the loop and control.
- RQ3 Deployment:** Scales to ~ 35 hosts, highlights key risks. **Requires supervision** (reliability limits).

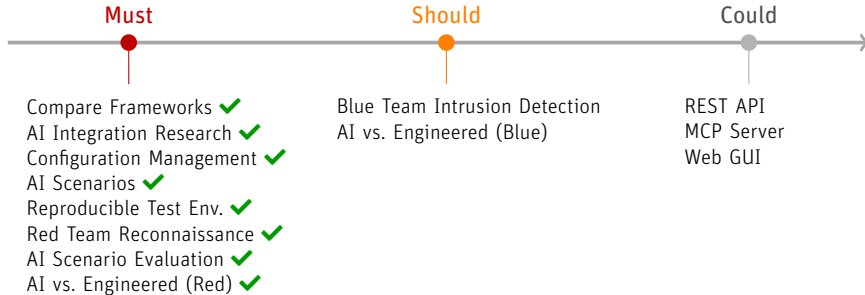
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Recommendations and Future Work

Recommendations

- **Match model tier to task** – frontier model for autonomous recon
- **Multi-agent** for smaller / self-hosted models
- **Hybrid pipeline**: static breadth + AI depth (+ RAG)
- **Human-in-the-loop** + retry on incomplete runs

Future work

- **Dynamic tool calling** – expose only relevant tools
- **Guardrails & sandboxing** – scope restriction for production
- **Robust HITL middleware**
- **Broader attack-chain** coverage beyond reconnaissance

Conclusion

Agentic AI complements deterministic enumeration

- ✓ Using NSAK directly as an **agentic harness** proved effective – the agent autonomously and interactively executes network discovery via LangChain tools.
- ✓ Scripted scenarios excel in correctness and completeness; the agent adds **reasoning and assessment**, limited by model capability.
- ✓ **Multi-agent decomposition** models to digest large context improves models by reducing context complexity and increasing consistency.
- ⚠ Hallucinations remain a limitation – **supervised operation with a human reviewer** is the proposed deployment model.

LLMs make cybersecurity use cases more accessible and cost-effective – with a human in the loop.

Personal Conclusion

Frank

- ContainerLab impressed me – simple to deploy and reconfigure, motivated a home cyber lab
- Enjoyed researching state-of-the-art models, papers and LangChain
- Prefer LLMs mainly for reasoning and recommendations
- Surprised how hard autonomous NSAK deployment in the NSLab was
- Deep respect for academic writing – keeping a consistent “red thread” is hard

Lukas

- Initially skeptical of the usefulness of AI
- We covered a lot of topics we learned at BFH
- A bit of naivety in the planning led to a larger scope
- Keep focusing on the goals and research questions
- AI is here to stay as another engineering tool

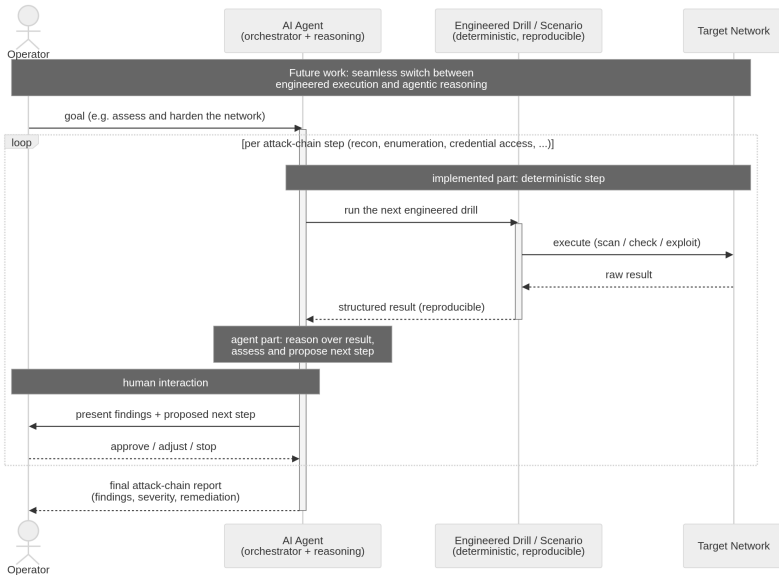


Figure:

Demo: Agentic Reconnaissance in Action